

GENE 525: Epigenomics and Chromatin Biology

Term: Fall 2026 | Instructor: Dr. Barbara McClintock (bmcclintock@genetics.univ.edu)

Course Description & Overview:

Mechanisms of DNA methylation, histone modification dynamics, 3D chromosomal architecture, and non-coding RNA gene regulation.

COURSE ASSIGNMENTS & WEIGHT DISTRIBUTION

Assignment Description	Weight	Due Module	Deliverable Format
ChIP-seq & ATAC-seq Bioinformatic Pipeline	30%	Week 04	Snakemake Pipeline & Analysis
Epigenetic Disease Mutation Report	30%	Week 08	Case Study & Variant Analysis
Chromatin Architecture Grant Proposal	35%	Week 11	NIH R01 Style Grant Proposal
Journal Article Critiques	5%	Week 06	Cell/Nature Review Critiques

WEEKLY TERM SCHEDULE & READING REQUIREMENTS

Timeline	Technical Topic Focus	Assigned Readings & Docs
Week 01	Module 01: DNA Methylation Mechanics & Islands	Allis et al. (Ch. 1–3); Bird (2002)
Week 02	Module 02: Histone Post-Translational Code	Jenuwein & Allis (2001)
Week 03	Module 03: Chromatin Remodeling (SWI/SNF)	Clapier & Cairns (2009)
Week 04	Module 04: Mapping Chromatin Accessibility: ATAC	Buenrostro et al. (2013); ENCODE Specs
Week 05	Module 05: 3D Genome Architecture: Hi-C & TADs	Lieberman-Aiden et al. (2009)
Week 06	Module 06: Non-Coding RNAs & X-Inactivation	Lee (2012); Rinn & Chang (2012)
Week 07	Module 07: Polycomb & Trithorax Group Memory	Schuettengruber et al. (2017)
Week 08	Module 08: Epigenetics in Cancer & Therapy	Baylin & Jones (2011)
Week 09	Module 09: Epigenetic Clock & Aging Biomarkers	Horvath (2013)
Week 10	Module 10: Environmental Transgenerational Traits	Heard & Martienssen (2014)
Week 11	Module 11: Final Grant Defense & Retrospective	Chromatin Biology Symposium